

Source folder: /Users/jonat/Documents/GitHub/master_thesis
Files in this part: 17

File: results/demo_run.json

```
{
  "experiment_id": "demo_run",
  "rounds": [
    {
      "round": 0,
      "train_loss": 11.710239797830582
    },
    {
      "round": 0,
      "train_loss": 11.6570865213871
    },
    {
      "round": 0,
      "train_loss": 11.580928921699524
    },
    {
      "round": 0,
      "train_loss": 11.605891764163971
    },
    {
      "round": 0,
      "train_loss": 11.562160149216652
    },
    {
      "round": 0,
      "train_loss": 11.499279126524925
    },
    {
      "round": 0,
      "train_loss": 11.629574418067932
    },
    {
      "round": 0,
      "train_loss": 11.632218688726425
    },
    {
      "round": 0,
      "train_loss": 11.711974412202835
    },
    {
      "round": 0,
      "train_loss": 11.652093887329102
    }
  ]
}
```

File: results/dp_1_alpha_0.5_dim_64_clients_100_seed_123.json

```
{
  "experiment_id": "dp_1_alpha_0.5_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T02:52:32.667743",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 1,
    "use_dp": true,
    "dp_sigma": 75.0625,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
  },
  "rounds": [
    {
```

```
"round": 1,
"train_loss": 38963.97423880543,
"test_metrics": {
  "mse": 26.465855584716795,
  "mae": 4.226308657836914,
  "samples": 20000,
  "NDCG@10": 0.04521064059975586,
  "Hit@10": 0.07,
  "Precision@10": 0.006999999999999999,
  "Recall@10": 0.0017224435186132148,
  "num_users_evaluated": 100
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 2,
  "train_loss": 40377.15169415534,
  "test_metrics": {
    "mse": 63.29663121948242,
    "mae": 6.364564102935791,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 3,
  "train_loss": 41567.93433565475,
  "test_metrics": {
    "mse": 125.221526171875,
    "mae": 8.902955809020996,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 43131.80167295857,
  "test_metrics": {
    "mse": 213.16070805664063,
    "mae": 11.645036456298827,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 44382.366078070045,
  "test_metrics": {
    "mse": 327.51094931640625,
    "mae": 14.392618478393555,
    "samples": 20000,
    "NDCG@10": 0.04050924333460803,
    "Hit@10": 0.03,
    "Precision@10": 0.0030000000000000005,
    "Recall@10": 0.002470653377630122,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
}
```

```
{
  "round": 6,
  "train_loss": 45507.749855667615,
  "test_metrics": {
    "mse": 459.7774859375,
    "mae": 17.018246719360352,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 46708.978485058025,
  "test_metrics": {
    "mse": 622.90454921875,
    "mae": 19.758551049804687,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 47813.48624437197,
  "test_metrics": {
    "mse": 812.9470924804688,
    "mae": 22.52230757446289,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 49770.981168050545,
  "test_metrics": {
    "mse": 1017.7248899414062,
    "mae": 25.38714585571289,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 50824.836797064934,
  "test_metrics": {
    "mse": 1264.3574362304687,
    "mae": 28.299958532714843,
    "samples": 20000,
    "NDCG@10": 0.038747436890763806,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.003423709984180309,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 1264.3574362304687,
  "mae": 28.299958532714843,
```

```

    "samples": 20000,
    "NDCG@10": 0.038747436890763806,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.003423709984180309,
    "num_users_evaluated": 100
  }
}

```

File: results/dp_1_alpha_0.5_dim_64_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,38963.97423880543,0.04521064059975586,0.07,0.006999999999999999,0.0017224435186132148,26.465855584716795,
4.226308657836914
2,40377.15169415534,,,,,63.29663121948242,6.364564102935791
3,41567.93433565475,,,,,125.221526171875,8.902955809020996
4,43131.80167295857,,,,,213.16070805664063,11.645036456298827
5,44382.366078070045,0.04050924333460803,0.03,0.0030000000000000005,0.002470653377630122,327.51094931640625
,14.392618478393555
6,45507.749855667615,,,,,459.7774859375,17.018246719360352
7,46708.978485058025,,,,,622.90454921875,19.758551049804687
8,47813.48624437197,,,,,812.9470924804688,22.52230757446289
9,49770.981168050545,,,,,1017.7248899414062,25.38714585571289
10,50824.836797064934,0.038747436890763806,0.05,0.005,0.003423709984180309,1264.3574362304687,28.2999585327
14843
FINAL,,0.038747436890763806,0.05,0.005,0.003423709984180309,1264.3574362304687,28.299958532714843

```

File: results/dp_1_alpha_0.5_dim_64_clients_100_seed_42.json

```

{
  "experiment_id": "dp_1_alpha_0.5_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T02:49:28.884828",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 1,
    "use_dp": true,
    "dp_sigma": 75.0625,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 38960.01599917924,
      "test_metrics": {
        "mse": 26.324150482177735,
        "mae": 4.210136761856079,
        "samples": 20000,
        "NDCG@10": 0.03961790233246403,
        "Hit@10": 0.07,
        "Precision@10": 0.008,
        "Recall@10": 0.003440055417135213,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 40141.44112942695,
      "test_metrics": {
        "mse": 62.564993969726565,
        "mae": 6.311017587280274,
        "samples": 20000
      }
    }
  ]
}

```

```
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 3,
    "train_loss": 41667.38061528948,
    "test_metrics": {
      "mse": 124.27513557128906,
      "mae": 8.879233229064942,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 42687.622308907325,
    "test_metrics": {
      "mse": 210.34479968261718,
      "mae": 11.537402191162109,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 5,
    "train_loss": 44143.33736494178,
    "test_metrics": {
      "mse": 317.18783852539065,
      "mae": 14.235216018676757,
      "samples": 20000,
      "NDCG@10": 0.04961254792901989,
      "Hit@10": 0.08,
      "Precision@10": 0.009000000000000001,
      "Recall@10": 0.005139105992047169,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 6,
    "train_loss": 45012.75422023619,
    "test_metrics": {
      "mse": 449.4102943359375,
      "mae": 16.89489171142578,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 7,
    "train_loss": 47048.08474836159,
    "test_metrics": {
      "mse": 616.3688580078125,
      "mae": 19.67854136657715,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
```

```
}
},
{
  "round": 8,
  "train_loss": 48080.34391334988,
  "test_metrics": {
    "mse": 778.6992239257812,
    "mae": 22.125241497802733,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 49151.029420506144,
  "test_metrics": {
    "mse": 983.3803299804688,
    "mae": 24.801917776489258,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 50547.670809926174,
  "test_metrics": {
    "mse": 1219.79353671875,
    "mae": 27.73233918762207,
    "samples": 20000,
    "NDCG@10": 0.06373179327591641,
    "Hit@10": 0.06,
    "Precision@10": 0.007000000000000001,
    "Recall@10": 0.0022059073478178253,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 1219.79353671875,
  "mae": 27.73233918762207,
  "samples": 20000,
  "NDCG@10": 0.06373179327591641,
  "Hit@10": 0.06,
  "Precision@10": 0.007000000000000001,
  "Recall@10": 0.0022059073478178253,
  "num_users_evaluated": 100
},
"attack_results": {
  "mia": {
    "train_accuracy": 1.0,
    "auc": 0.48584999999999995,
    "accuracy": 0.51,
    "true_positive_rate": 0.62,
    "false_positive_rate": 0.6
  },
  "inversion": {
    "top_k_accuracy": 0.0,
    "top_k": 10,
    "num_users_evaluated": 50
  }
}
}
```

File: results/dp_1_alpha_0.5_dim_64_clients_100_seed_42_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,38960.01599917924,0.03961790233246403,0.07,0.008,0.003440055417135213,26.324150482177735,4.21013676185607
9
2,40141.44112942695,,,,,62.564993969726565,6.311017587280274
3,41667.38061528948,,,,,124.27513557128906,8.879233229064942
4,42687.622308907325,,,,,210.34479968261718,11.537402191162109
5,44143.33736494178,0.04961254792901989,0.08,0.009000000000000001,0.005139105992047169,317.18783852539065,1
4.235216018676757
6,45012.75422023619,,,,,449.4102943359375,16.89489171142578
7,47048.08474836159,,,,,616.3688580078125,19.67854136657715
8,48080.34391334988,,,,,778.6992239257812,22.125241497802733
9,49151.029420506144,,,,,983.3803299804688,24.801917776489258
10,50547.670809926174,0.06373179327591641,0.06,0.007000000000000001,0.0022059073478178253,1219.79353671875,
27.73233918762207
FINAL,,0.06373179327591641,0.06,0.007000000000000001,0.0022059073478178253,1219.79353671875,27.732339187622
07
```

File: results/dp_1_alpha_0.5_dim_64_clients_100_seed_456.json

```
{
  "experiment_id": "dp_1_alpha_0.5_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T02:55:35.345975",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 1,
    "use_dp": true,
    "dp_sigma": 75.0625,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 38904.826062472566,
      "test_metrics": {
        "mse": 25.98537597351074,
        "mae": 4.186658390045166,
        "samples": 20000,
        "NDCG@10": 0.040225689933686935,
        "Hit@10": 0.07,
        "Precision@10": 0.008,
        "Recall@10": 0.0035781231096256656,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 40290.495299244,
      "test_metrics": {
        "mse": 62.07684714355469,
        "mae": 6.308922241210937,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 3,
```

```
"train_loss": 41192.94593482481,
"test_metrics": {
  "mse": 120.73841142578125,
  "mae": 8.752308038330078,
  "samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 4,
  "train_loss": 42551.16011562006,
  "test_metrics": {
    "mse": 206.5189384765625,
    "mae": 11.415081701660156,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 43977.35872124249,
  "test_metrics": {
    "mse": 312.61522465820315,
    "mae": 14.057603115844726,
    "samples": 20000,
    "NDCG@10": 0.04299470147220183,
    "Hit@10": 0.06,
    "Precision@10": 0.0070000000000000001,
    "Recall@10": 0.005984292189093961,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 45112.42385186509,
  "test_metrics": {
    "mse": 446.0535176757812,
    "mae": 16.764670126342775,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 46971.01330695465,
  "test_metrics": {
    "mse": 600.2338583007812,
    "mae": 19.36639291381836,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 47725.49999426106,
  "test_metrics": {
    "mse": 780.3972463867187,
    "mae": 22.106222232055664,
```



```

    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 49063.3764904598,
  "test_metrics": {
    "mse": 976.159595703125,
    "mae": 24.826724716186522,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 50430.77642351835,
  "test_metrics": {
    "mse": 1200.692373046875,
    "mae": 27.474232513427733,
    "samples": 20000,
    "NDCG@10": 0.03762802832552227,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.002745130885374788,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 1200.692373046875,
  "mae": 27.474232513427733,
  "samples": 20000,
  "NDCG@10": 0.03762802832552227,
  "Hit@10": 0.05,
  "Precision@10": 0.005,
  "Recall@10": 0.002745130885374788,
  "num_users_evaluated": 100
}
}

```

File: results/dp_1_alpha_0.5_dim_64_clients_100_seed_456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,38904.826062472566,0.040225689933686935,0.07,0.008,0.0035781231096256656,25.98537597351074,4.186658390045
166
2,40290.495299244,,,,,62.07684714355469,6.308922241210937
3,41192.94593482481,,,,,120.73841142578125,8.752308038330078
4,42551.16011562006,,,,,206.5189384765625,11.415081701660156
5,43977.35872124249,0.04299470147220183,0.06,0.0070000000000000001,0.005984292189093961,312.61522465820315,1
4.057603115844726
6,45112.42385186509,,,,,446.0535176757812,16.764670126342775
7,46971.01330695465,,,,,600.2338583007812,19.36639291381836
8,47725.49999426106,,,,,780.3972463867187,22.106222232055664
9,49063.3764904598,,,,,976.159595703125,24.826724716186522
10,50430.77642351835,0.03762802832552227,0.05,0.005,0.002745130885374788,1200.692373046875,27.4742325134277
33
FINAL,,0.03762802832552227,0.05,0.005,0.002745130885374788,1200.692373046875,27.474232513427733

```

File: results/dp_2_alpha_0.5_dim_64_clients_100_seed_123.json

```

{
  "experiment_id": "dp_2_alpha_0.5_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T02:42:57.478541",

```

```
"config": {
  "num_users": 943,
  "num_items": 1682,
  "embedding_dim": 64,
  "num_clients": 100,
  "alpha": 0.5,
  "dp_epsilon": 2,
  "use_dp": true,
  "dp_sigma": 40.7046875,
  "dp_clip_norm": 1.0,
  "num_rounds": 10,
  "local_epochs": 3,
  "learning_rate": 0.01,
  "batch_size": 32,
  "seed": 123
},
"rounds": [
  {
    "round": 1,
    "train_loss": 3384.70252605272,
    "test_metrics": {
      "mse": 14.924642190551758,
      "mae": 3.540842860031128,
      "samples": 20000,
      "NDCG@10": 0.04847351432133537,
      "Hit@10": 0.07,
      "Precision@10": 0.007000000000000001,
      "Recall@10": 0.0016603465518112087,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 2,
    "train_loss": 3507.607729213016,
    "test_metrics": {
      "mse": 18.18878537902832,
      "mae": 3.677388525772095,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 3,
    "train_loss": 3610.2767853764985,
    "test_metrics": {
      "mse": 23.594636837768554,
      "mae": 4.026535112762451,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 3746.363727664809,
    "test_metrics": {
      "mse": 31.289648999023438,
      "mae": 4.534889735412598,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
],
```

```
{
  "round": 5,
  "train_loss": 3856.7523849416866,
  "test_metrics": {
    "mse": 41.48843837890625,
    "mae": 5.180301279449463,
    "samples": 20000,
    "NDCG@10": 0.03870433732912934,
    "Hit@10": 0.03,
    "Precision@10": 0.0030000000000000005,
    "Recall@10": 0.002470653377630122,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 3951.9481274582445,
  "test_metrics": {
    "mse": 53.106045880126956,
    "mae": 5.82088099822998,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 4058.3124878553817,
  "test_metrics": {
    "mse": 67.12436173095703,
    "mae": 6.513197617340088,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 4151.770754174233,
  "test_metrics": {
    "mse": 83.58081507568359,
    "mae": 7.2610292114257815,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 4320.006720739383,
  "test_metrics": {
    "mse": 101.15629572753906,
    "mae": 8.025469793701172,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 4409.5620183469255,
  "test_metrics": {
```

```

        "mse": 122.77108743896484,
        "mae": 8.838040160369873,
        "samples": 20000,
        "NDCG@10": 0.03996836927225674,
        "Hit@10": 0.05,
        "Precision@10": 0.005,
        "Recall@10": 0.003423709984180309,
        "num_users_evaluated": 100
    },
    "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
    }
}
],
"final_metrics": {
    "mse": 122.77108743896484,
    "mae": 8.838040160369873,
    "samples": 20000,
    "NDCG@10": 0.03996836927225674,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.003423709984180309,
    "num_users_evaluated": 100
}
}

```

File: results/dp_2_alpha_0.5_dim_64_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,3384.70252605272,0.04847351432133537,0.07,0.007000000000000001,0.0016603465518112087,14.924642190551758,3
.540842860031128
2,3507.607729213016,,,,,18.18878537902832,3.677388525772095
3,3610.2767853764985,,,,,23.594636837768554,4.026535112762451
4,3746.363727664809,,,,,31.289648999023438,4.534889735412598
5,3856.7523849416866,0.03870433732912934,0.03,0.0030000000000000005,0.002470653377630122,41.48843837890625,
5.180301279449463
6,3951.9481274582445,,,,,53.106045880126956,5.82088099822998
7,4058.3124878553817,,,,,67.12436173095703,6.513197617340088
8,4151.770754174233,,,,,83.58081507568359,7.2610292114257815
9,4320.006720739383,,,,,101.15629572753906,8.025469793701172
10,4409.5620183469255,0.03996836927225674,0.05,0.005,0.003423709984180309,122.77108743896484,8.838040160369
873
FINAL,,0.03996836927225674,0.05,0.005,0.003423709984180309,122.77108743896484,8.838040160369873

```

File: results/dp_2_alpha_0.5_dim_64_clients_100_seed_42.json

```

{
  "experiment_id": "dp_2_alpha_0.5_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T02:38:48.431009",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 2,
    "use_dp": true,
    "dp_sigma": 40.7046875,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 3386.0751333259045,
      "test_metrics": {
        "mse": 14.867079779052734,
        "mae": 3.5347402950286866,
        "samples": 20000,

```

```
"NDCG@10": 0.03971305784950692,
"Hit@10": 0.06,
"Precision@10": 0.006999999999999999,
"Recall@10": 0.0027202326042891443,
"num_users_evaluated": 100
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 2,
  "train_loss": 3488.49369102608,
  "test_metrics": {
    "mse": 17.96334690246582,
    "mae": 3.6511829357147216,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 3,
  "train_loss": 3620.4820652038243,
  "test_metrics": {
    "mse": 23.45289249267578,
    "mae": 4.01370072479248,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 3707.3488921543467,
  "test_metrics": {
    "mse": 31.128795666503905,
    "mae": 4.532878741455078,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 3833.9187078650975,
  "test_metrics": {
    "mse": 40.42511262817383,
    "mae": 5.121992610931397,
    "samples": 20000,
    "NDCG@10": 0.04759520380012195,
    "Hit@10": 0.08,
    "Precision@10": 0.009000000000000001,
    "Recall@10": 0.005139105992047169,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 3908.0879557490157,
  "test_metrics": {
    "mse": 51.90749056396484,
    "mae": 5.775124337005615,
```

```
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 4082.852249748841,
  "test_metrics": {
    "mse": 66.30026739501953,
    "mae": 6.4860578056335445,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 4174.175201685227,
  "test_metrics": {
    "mse": 80.15156484375,
    "mae": 7.126118628692627,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 4266.877559568634,
  "test_metrics": {
    "mse": 98.07662218017578,
    "mae": 7.861813136291504,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 4389.6386369329675,
  "test_metrics": {
    "mse": 118.63443210449219,
    "mae": 8.673735711669922,
    "samples": 20000,
    "NDCG@10": 0.05896409118693234,
    "Hit@10": 0.07,
    "Precision@10": 0.007000000000000001,
    "Recall@10": 0.0021049281556513627,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 118.63443210449219,
  "mae": 8.673735711669922,
  "samples": 20000,
  "NDCG@10": 0.05896409118693234,
  "Hit@10": 0.07,
  "Precision@10": 0.007000000000000001,
  "Recall@10": 0.0021049281556513627,
  "num_users_evaluated": 100
}
```

```

},
"attack_results": {
  "mia": {
    "train_accuracy": 1.0,
    "auc": 0.5004,
    "accuracy": 0.5,
    "true_positive_rate": 0.64,
    "false_positive_rate": 0.64
  },
  "inversion": {
    "top_k_accuracy": 0.0,
    "top_k": 10,
    "num_users_evaluated": 50
  }
}
}
}

```

File: results/dp_2_alpha_0.5_dim_64_clients_100_seed_42_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,3386.0751333259045,0.03971305784950692,0.06,0.006999999999999999,0.0027202326042891443,14.867079779052734
,3.5347402950286866
2,3488.49369102608,,,,,17.96334690246582,3.6511829357147216
3,3620.4820652038243,,,,,23.45289249267578,4.01370072479248
4,3707.3488921543467,,,,,31.128795666503905,4.532878741455078
5,3833.9187078650975,0.04759520380012195,0.08,0.009000000000000001,0.005139105992047169,40.42511262817383,5
.121992610931397
6,3908.0879557490157,,,,,51.90749056396484,5.775124337005615
7,4082.852249748841,,,,,66.30026739501953,6.4860578056335445
8,4174.175201685227,,,,,80.15156484375,7.126118628692627
9,4266.877559568634,,,,,98.07662218017578,7.861813136291504
10,4389.6386369329675,0.05896409118693234,0.07,0.007000000000000001,0.0021049281556513627,118.6344321044921
9,8.673735711669922
FINAL,,0.05896409118693234,0.07,0.007000000000000001,0.0021049281556513627,118.63443210449219,8.67373571166
9922

```

File: results/dp_2_alpha_0.5_dim_64_clients_100_seed_456.json

```

{
  "experiment_id": "dp_2_alpha_0.5_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T02:46:22.816395",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 2,
    "use_dp": true,
    "dp_sigma": 40.7046875,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 3379.9243578834735,
      "test_metrics": {
        "mse": 14.82579482421875,
        "mae": 3.5305438163757326,
        "samples": 20000,
        "NDCG@10": 0.04543685807395247,
        "Hit@10": 0.09,
        "Precision@10": 0.009000000000000001,
        "Recall@10": 0.003351877595037266,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    }
  ]
}

```

```
}
},
{
  "round": 2,
  "train_loss": 3499.3596683412675,
  "test_metrics": {
    "mse": 18.088604051208495,
    "mae": 3.6784022804260252,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 3,
  "train_loss": 3579.559214961313,
  "test_metrics": {
    "mse": 23.290728747558592,
    "mae": 4.014856020355225,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 3695.6306296415255,
  "test_metrics": {
    "mse": 30.755915740966795,
    "mae": 4.5036157135009764,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 3817.37629137021,
  "test_metrics": {
    "mse": 40.1239346496582,
    "mae": 5.095039193725586,
    "samples": 20000,
    "NDCG@10": 0.04656793552925941,
    "Hit@10": 0.05,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.0026509588557606283,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 3917.4636466761917,
  "test_metrics": {
    "mse": 51.799483715820315,
    "mae": 5.744916427612305,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
```



```

    "train_loss": 4077.879251762293,
    "test_metrics": {
      "mse": 65.4899167602539,
      "mae": 6.441198127746582,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 8,
    "train_loss": 4145.120165642374,
    "test_metrics": {
      "mse": 81.12905133056641,
      "mae": 7.158955921173096,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 9,
    "train_loss": 4258.075307701938,
    "test_metrics": {
      "mse": 97.96482468261719,
      "mae": 7.883077192687988,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 10,
    "train_loss": 4376.930281931637,
    "test_metrics": {
      "mse": 117.72294920654296,
      "mae": 8.617804049682617,
      "samples": 20000,
      "NDCG@10": 0.03776224559152501,
      "Hit@10": 0.05,
      "Precision@10": 0.005,
      "Recall@10": 0.0023605155007594032,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
],
"final_metrics": {
  "mse": 117.72294920654296,
  "mae": 8.617804049682617,
  "samples": 20000,
  "NDCG@10": 0.03776224559152501,
  "Hit@10": 0.05,
  "Precision@10": 0.005,
  "Recall@10": 0.0023605155007594032,
  "num_users_evaluated": 100
}
}

```

File: results/dp_2_alpha_0.5_dim_64_clients_100_seed_456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,3379.9243578834735,0.04543685807395247,0.09,0.009000000000000001,0.003351877595037266,14.82579482421875,3
.5305438163757326
2,3499.3596683412675,,,,,18.088604051208495,3.6784022804260252

```

```
3,3579.559214961313,,,,,23.290728747558592,4.014856020355225
4,3695.6306296415255,,,,,30.755915740966795,4.5036157135009764
5,3817.37629137021,0.04656793552925941,0.05,0.0060000000000000001,0.0026509588557606283,40.1239346496582,5.0
95039193725586
6,3917.4636466761917,,,,,51.799483715820315,5.744916427612305
7,4077.879251762293,,,,,65.4899167602539,6.441198127746582
8,4145.120165642374,,,,,81.12905133056641,7.158955921173096
9,4258.075307701938,,,,,97.96482468261719,7.883077192687988
10,4376.930281931637,0.03776224559152501,0.05,0.005,0.0023605155007594032,117.72294920654296,8.617804049682
617
FINAL,,0.03776224559152501,0.05,0.005,0.0023605155007594032,117.72294920654296,8.617804049682617
```

File: results/dp_4_alpha_0.5_dim_64_clients_100_seed_123.json

```
{
  "experiment_id": "dp_4_alpha_0.5_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T02:31:10.230233",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 4,
    "use_dp": true,
    "dp_sigma": 20.40234375,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 227.39276497234422,
      "test_metrics": {
        "mse": 13.800793028259278,
        "mae": 3.522211731719971,
        "samples": 20000,
        "NDCG@10": 0.05185266953566699,
        "Hit@10": 0.06,
        "Precision@10": 0.0060000000000000001,
        "Recall@10": 0.0011885987128906943,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 235.41270588546533,
      "test_metrics": {
        "mse": 14.025225309753418,
        "mae": 3.5217144886016847,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 3,
      "train_loss": 241.86451076223912,
      "test_metrics": {
        "mse": 14.380878532409668,
        "mae": 3.52511671333313,
        "samples": 20000
      },
      "aggregation": {
```

```
    "num_clients": 100,
    "total_samples": 80000
  },
  {
    "round": 4,
    "train_loss": 250.65472324378024,
    "test_metrics": {
      "mse": 14.879275038146973,
      "mae": 3.532377660369873,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 5,
    "train_loss": 258.0789284126659,
    "test_metrics": {
      "mse": 15.601707807922363,
      "mae": 3.5585600902557375,
      "samples": 20000,
      "NDCG@10": 0.043454917351544095,
      "Hit@10": 0.05,
      "Precision@10": 0.005,
      "Recall@10": 0.0038039867109634553,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 6,
    "train_loss": 263.62437503811003,
    "test_metrics": {
      "mse": 16.389767547607423,
      "mae": 3.5958754760742186,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 7,
    "train_loss": 270.9868775376554,
    "test_metrics": {
      "mse": 17.26231935119629,
      "mae": 3.630116690826416,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 8,
    "train_loss": 276.4068795810124,
    "test_metrics": {
      "mse": 18.307551095581054,
      "mae": 3.6863674404144287,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
},
```

```

{
  "round": 9,
  "train_loss": 286.78034017667227,
  "test_metrics": {
    "mse": 19.379044264221193,
    "mae": 3.753527548599243,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 292.1784286954096,
  "test_metrics": {
    "mse": 20.81909543457031,
    "mae": 3.850083127212524,
    "samples": 20000,
    "NDCG@10": 0.048026363096703564,
    "Hit@10": 0.06,
    "Precision@10": 0.006,
    "Recall@10": 0.00366180522275547,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
},
"final_metrics": {
  "mse": 20.81909543457031,
  "mae": 3.850083127212524,
  "samples": 20000,
  "NDCG@10": 0.048026363096703564,
  "Hit@10": 0.06,
  "Precision@10": 0.006,
  "Recall@10": 0.00366180522275547,
  "num_users_evaluated": 100
}
}

```

File: results/dp_4_alpha_0.5_dim_64_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,227.39276497234422,0.05185266953566699,0.06,0.006000000000000001,0.0011885987128906943,13.800793028259278
,3.522211731719971
2,235.41270588546533,,,,,14.025225309753418,3.5217144886016847
3,241.86451076223912,,,,,14.380878532409668,3.52511671333313
4,250.65472324378024,,,,,14.879275038146973,3.532377660369873
5,258.0789284126659,0.043454917351544095,0.05,0.005,0.0038039867109634553,15.601707807922363,3.558560090255
7375
6,263.62437503811003,,,,,16.389767547607423,3.5958754760742186
7,270.9868775376554,,,,,17.26231935119629,3.630116690826416
8,276.4068795810124,,,,,18.307551095581054,3.6863674404144287
9,286.78034017667227,,,,,19.379044264221193,3.753527548599243
10,292.1784286954096,0.048026363096703564,0.06,0.006,0.00366180522275547,20.81909543457031,3.8500831272125
24
FINAL,,0.048026363096703564,0.06,0.006,0.00366180522275547,20.81909543457031,3.850083127212524

```

File: results/dp_4_alpha_0.5_dim_64_clients_100_seed_42.json

```

{
  "experiment_id": "dp_4_alpha_0.5_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T02:26:43.406751",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 4,

```

```
"use_dp": true,
"dp_sigma": 20.40234375,
"dp_clip_norm": 1.0,
"num_rounds": 10,
"local_epochs": 3,
"learning_rate": 0.01,
"batch_size": 32,
"seed": 42
},
"rounds": [
{
  "round": 1,
  "train_loss": 228.00020887873922,
  "test_metrics": {
    "mse": 13.7829908203125,
    "mae": 3.5202133369445803,
    "samples": 20000,
    "NDCG@10": 0.06573527998901081,
    "Hit@10": 0.11,
    "Precision@10": 0.011000000000000001,
    "Recall@10": 0.005121197136832625,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 2,
  "train_loss": 234.42883935513137,
  "test_metrics": {
    "mse": 13.968070623779298,
    "mae": 3.513886647796631,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 3,
  "train_loss": 242.85100448068195,
  "test_metrics": {
    "mse": 14.35976184539795,
    "mae": 3.523405859375,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 248.08766878346557,
  "test_metrics": {
    "mse": 14.908065705871582,
    "mae": 3.540966688156128,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 256.18368029690765,
  "test_metrics": {
    "mse": 15.50926480102539,
    "mae": 3.5578789264678954,
    "samples": 20000,
```

```
"NDCG@10": 0.04759706688923452,
"Hit@10": 0.07,
"Precision@10": 0.008,
"Recall@10": 0.00414447614931606,
"num_users_evaluated": 100
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 6,
  "train_loss": 260.72376313686743,
  "test_metrics": {
    "mse": 16.2442454788208,
    "mae": 3.5853822803497315,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 271.36841570603195,
  "test_metrics": {
    "mse": 17.139047613525392,
    "mae": 3.619733070373535,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 277.6128170244431,
  "test_metrics": {
    "mse": 17.961268449401857,
    "mae": 3.6585649604797363,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 283.44371966813014,
  "test_metrics": {
    "mse": 19.153935430908202,
    "mae": 3.7339311351776123,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 291.8478941332176,
  "test_metrics": {
    "mse": 20.485859423828124,
    "mae": 3.8241310848236085,
    "samples": 20000,
    "NDCG@10": 0.05901090278606798,
    "Hit@10": 0.07,
    "Precision@10": 0.006999999999999999,
    "Recall@10": 0.0021080536440089186,
```

```

        "num_users_evaluated": 100
    },
    "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
    }
},
"final_metrics": {
    "mse": 20.485859423828124,
    "mae": 3.8241310848236085,
    "samples": 20000,
    "NDCG@10": 0.05901090278606798,
    "Hit@10": 0.07,
    "Precision@10": 0.006999999999999999,
    "Recall@10": 0.0021080536440089186,
    "num_users_evaluated": 100
},
"attack_results": {
    "mia": {
        "train_accuracy": 1.0,
        "auc": 0.5024500000000001,
        "accuracy": 0.505,
        "true_positive_rate": 0.61,
        "false_positive_rate": 0.6
    },
    "inversion": {
        "top_k_accuracy": 0.0,
        "top_k": 10,
        "num_users_evaluated": 50
    }
}
}
}

```

File: results/dp_4_alpha_0.5_dim_64_clients_100_seed_42_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,228.00020887873922,0.06573527998901081,0.11,0.011000000000000001,0.005121197136832625,13.7829908203125,3.5202133369445803
2,234.42883935513137,,,,,13.968070623779298,3.513886647796631
3,242.85100448068195,,,,,14.35976184539795,3.523405859375
4,248.08766878346557,,,,,14.908065705871582,3.540966688156128
5,256.18368029690765,0.04759706688923452,0.07,0.008,0.00414447614931606,15.50926480102539,3.557878926467895
6,260.72376313686743,,,,,16.2442454788208,3.5853822803497315
7,271.36841570603195,,,,,17.139047613525392,3.619733070373535
8,277.6128170244431,,,,,17.961268449401857,3.6585649604797363
9,283.44371966813014,,,,,19.153935430908202,3.7339311351776123
10,291.8478941332176,0.05901090278606798,0.07,0.006999999999999999,0.0021080536440089186,20.485859423828124,3.8241310848236085
FINAL,,0.05901090278606798,0.07,0.006999999999999999,0.0021080536440089186,20.485859423828124,3.8241310848236085

```